ML4HOR: Machine Learning for Health Outcomes with R

Full Program, September 2026 Cohort

Imad EL BADISY, PhD

1 Course Overview

Title: Machine Learning for Health Outcomes with R
Format: Online, cohort-based
Duration: 2 weeks, 8 sessions, 2 hours per session, 16 contact hours
Dates: 21/09/2026 – 04/10/2026
Days: Monday, Wednesday, Friday (18h00 to 20h00), Sunday (10h00 to 12h00)
Time: Morocco time
Price: 150€ (1500 MAD)
Instructor: Imad EL BADISY, PhD
Contact: elbadisyimad@gmail.com | elbadisyimad.com

1.1 Course Description

`ML4HOR` provides a practical introduction to machine learning for health outcomes research using R. The course covers the complete supervised learning workflow and explores clustering as a final extension for patient phenotyping, subgroup discovery, and exploratory health-data analysis.

The course emphasizes validation, data leakage prevention, appropriate performance metrics, calibration, interpretability, clinical relevance, and reproducible reporting.

1.2 Target Audience

Researchers, PhD students, clinicians, public health professionals, and data analysts. Basic R and statistical literacy recommended.

1.3 Prerequisites

Basic familiarity with R (as covered in `BIOSTATR`) is recommended but not mandatory.

1.4 Learning Objectives

By the end of `ML4HOR`, participants will be able to:

1. formulate machine learning questions in health outcomes research;
2. prepare tabular health datasets for ML analysis;
3. fit supervised learners;
4. evaluate models using resampling-based validation;
5. tune and compare learners fairly;
6. assess discrimination, calibration, and clinical usefulness;
7. interpret ML models using global and local explanation methods;
8. apply clustering for patient phenotyping and subgroup discovery;
9. distinguish prediction, clustering, and causal estimation;
10. produce a reproducible ML report with Quarto.

2 Detailed Schedule

Table 1: `ML4HOR` detailed schedule

No.	Week	Date	Time	Topic	Main Content
1	1	21/09/2026 (Mon)	18h00-20h00	Introduction to ML for Health Outcomes & R/Quarto Workflow	Prediction vs explanation, outcome types, supervised vs unsupervised learning, leakage risks, data preparation
2	1	23/09/2026 (Wed)	18h00-20h00	Resampling and Performance Metrics	Train/test split, cross-validation, bootstrap, RMSE, MAE, AUC, F1, log-loss, Brier score
3	1	25/09/2026 (Fri)	18h00-20h00	Linear and Regularized Models	GLM, logistic regression, ridge, lasso, elastic net, shrinkage, high-dimensional predictors
4	1	27/09/2026 (Sun)	10h00-12h00	Tree-Based Models, Random Forests, Boosting & Ensembles	CART, random forests, variable importance, XGBoost, LightGBM, SVM, kNN, stacking
5	2	28/09/2026 (Mon)	18h00-20h00	Hyperparameter Tuning, Model Selection & Benchmarking	Grid/random search, nested CV, optimism, fair comparison, benchmark tables
6	2	30/09/2026 (Wed)	18h00-20h00	Calibration and Clinical Utility	Calibration curves, discrimination vs calibration, thresholds, sensitivity, specificity, ROC-AUC
7	2	02/10/2026 (Fri)	18h00-20h00	Model Interpretation, Explainability & Clustering for Phenotyping	Permutation importance, PDP, ICE, ALE, SHAP, k-means, hierarchical clustering, cluster profiling
8	2	04/10/2026 (Sun)	10h00-12h00	End-to-End ML Pipeline and Mini-Project	Full workflow: prepare, fit or cluster, evaluate, tune/compare, calibrate, interpret, report

3 Course Materials

- Slides and self-contained notebooks distributed before each session
- R scripts, datasets, and exercises (shared repository)
- Exercises reviewed and corrected during sessions

4 Assessment

Each course ends with a short reproducible Quarto report or mini-project, presented and discussed in Session 8.

5 About the Instructor

Imad EL BADISY, PhD, is a biostatistician working on statistical and machine learning methods for health data, with a focus on survival analysis. Full profile, publications, and R packages: elbadisyimad.com

6 Policies

Language: Teaching in French, with English for statistical and methodological terminology.
Software: R ($\geq$ 4.4) and RStudio required.
Attendance: Full attendance at all 8 sessions is recommended given the cumulative structure.